MD TAMZID HOSSEN OMI

IoT and Robotics Engineer | *Embedded Systems* | *AI & Software Engineer*

+8801882131123 | tamzid.omi@gmail.com | Github: tamzid16 | Linkedin: MD Tamzid Hossen Omi | Dhaka, Bangladesh

PROFESSIONAL SUMMARY

B.Sc. graduate in IoT and Robotics Engineering (2026) with a research-oriented profile covering embedded firmware, autonomous robotics, IoT-cloud integration and applied AI. Co-author of an IEEE conference paper on an autonomous disaster response rover and researcher on Agri-RAG, a multimodal retrieval-augmented framework for crop disease diagnosis. Has built and publicly released 20+ repositories spanning RAG platforms, Flutter mobile apps, FastAPI services and C/C++ embedded systems. A fresher with demonstrated initiative across project, research and self-driven project work.

TECHNICAL SKILLS

Languages: Python, C/C++ (Embedded), JavaScript (ES6+), TypeScript, Dart, SQL, LaTeX

IoT and Embedded: Arduino, Raspberry Pi, ESP32, ESP8266, MQTT, I2C, UART, SPI, RF Telemetry, Sensor and Actuator Integration

Robotics: Autonomous Navigation, PID Control, Multi-Sensor Fusion, Obstacle Avoidance, State Machine Design, Voice and RF Control

AI and Machine Learning: RAG Pipelines, Vision-Language Models (VLMs), LLM Integration, CNN Classification, FAISS, pgvector, Ollama, OpenCV, Scikit-learn, Pandas, NumPy

Web, Mobile and Databases: Next.js, FastAPI, Node.js, Flutter (Riverpod, Clean Architecture), Prisma ORM, PostgreSQL, MySQL, SQLite, Firebase, REST API Design

Developer Tools: Git, GitHub, Linux (Ubuntu), Android Studio, Visual Studio, VS Code, Technical Documentation

Additional: Digital Marketing (Meta Ads Manager), Graphic Design (Photoshop, Canva), Video Editing, Microsoft Office Suite

RESEARCH EXPERIENCE

JuteRAG — Vision-RAG Pipeline for Bangla Jute Disease Diagnosis | *2026*

- Built a pipeline combining DenseNet121 CNN classification, FAISS-indexed vector retrieval and LLM-generated Bangla advisory output for jute crop disease support.

WatchRover — IoT Autonomous Disaster-Response Robot | *Co-Author* | *IEEE QPAIN International Conference, CUET* | *2026*

- Developed ESP32 rover firmware fusing ultrasonic, IMU and IR sensor data within an adaptive navigation state machine; implemented RF telemetry for live monitoring and manual override.

SELECTED PROJECTS

KnowledgeOS and RAGBench — AI/RAG Engineering Platforms | *Next.js, FastAPI, PostgreSQL/pgvector, Ollama*

- KnowledgeOS: Local-first RAG platform with hybrid search, reranking, streaming chat, source citations and a built-in evaluation suite.
- RAGBench: Controlled RAG evaluation lab enabling side-by-side run comparisons across chunking, retrieval and prompt strategies.

RepoPilot AI and AgentDesk | *FastAPI, tree-sitter, pgvector, Next.js*

- RepoPilot AI: Code-aware RAG with tree-sitter AST chunking and line-level citations for codebase Q&A.
- AgentDesk: AI agent execution and observability platform with a 26-test automated backend test suite.

GHURI — Flutter and IoT Travel Safety Platform | *Flutter, Dart, FastAPI, GPS, SOS Alerting*

- Cross-platform travel app with real-time GPS tracking, hardware-integrated SOS alerts, booking and payment flows, backed by an IoT communication layer for emergency broadcasts.

IoT Environmental Monitoring and Smart Irrigation Systems | *Arduino, Raspberry Pi, MQTT, Python, Embedded C* | *Academic*

- Multi-node sensor network with cloud aggregation and threshold alerting; automated irrigation from real-time soil moisture feedback with edge-based relay control.

Fish Recommendation System — Smart Farming ML Module | *Python, Scikit-learn, Pandas, NumPy* | *Academic*

- Supervised classifier recommending aquaculture fish species from water quality, temperature and geographic parameters; full pipeline from sensor input through feature engineering to inference.

EDUCATION

B.Sc. in Internet of Things and Robotics Engineering | *University of Frontier Technology, Bangladesh (UFTB)* | *2026*

Higher Secondary Certificate (HSC) | *Rajarbagh Police Lines School and College, Dhaka* | *2020*

Secondary School Certificate (SSC) | *Motijheel Government Boys' High School, Dhaka* | *2018*

LEADERSHIP AND ACTIVITIES

- Joint Secretary, UFTB Robotics Club — Contributed to organising club events and student engineering initiatives.
- Executive Member, Robofusion IRE 1.0 — Assisted in planning the national level robotics event.

ACHIEVEMENTS AND AWARDS

- IEEE QPAIN International Conference 2026 (CUET) — Co-authored WatchRover.
- UIU CSE Fest 2025 — Team Leader, "Red Crew", Project Show.
- National Innovation Competition 2024 (TechnoCrats V.2) — Finalist.
- Bangladesh Environment Olympiad — Green Earth Quest, Round Champion.
- Grameenphone Accelerator Program — Tech Entrepreneurship Track, Savar Region.

CERTIFICATIONS AND TRAINING

- Cyber Security Training — Creative IT Institute, Dhaka
- Python Programming Fundamentals — Songjog Connecting People Foundation
- 21st Century Employability Skilling Program (Advanced) — Wadhwani Foundation

LANGUAGES

Bangla: Native
English: Professional Working Proficiency

REFERENCE

Md. Toukir Ahmed
Assistant Professor
Department of Internet of Things and Robotics Engineering
University of Frontier Technology, Bangladesh
Email: toukir0001@uftb.ac.bd
Phone: +8801738039128

Saurav Chandra Das
Lecturer
Department of Internet of Things and Robotics Engineering
University of Frontier Technology, Bangladesh
Email: saurav0001@uftb.ac.bd
Phone: +88015434695